PrivLens: Privacy Evidence & Metadata Inventory

Empirical PII Discovery, RoPA Governance Audit & Benchmark Report | Generated: October 31, 2026 • 23:17 UTC

LEGAL BOUNDARY & AUDIT NOTICE: This technical report was compiled through automated syntactic discovery and column profiling. It represents factual technical evidence intended for privacy steward and DPO review. It **does NOT** constitute an automated lawful-basis decision under GDPR/CCPA, nor does it deduce universal cross-border transfer topologies.

1. Executive Technical Summary

Metric Name	Count / Value	Context & Classification Scope
Total Fields Scanned	**44**	Structured CSV spreadsheets and SQLite database tables
Candidate PII Discovered	**31**	Personal data fields across 5 predefined data classes
Non-PII Columns	13	Technical keys, surrogate IDs, timestamps, status flags
Governance & Evidence Gaps	**15**	Unregistered shadow data, missing purposes, or missing retention

Distribution Across 5 Predefined Data Classes

Data Class Name	Candidate Fields	Scope & Subtypes Observed
Personal identity (PID)	6	Name, date of birth, gender, age
Contact information (CNT)	4	Email address, phone number
Government identifiers (GID)	7	Aadhaar, PAN, passport, voter ID, driving licence
Financial information (FIN)	8	Bank account number, IFSC, card number, UPI ID
Location / address (LOC)	6	Postal address, PIN code, geo-coordinates

2. Structured Technical Evidence Inventory

Target Asset / Column	Discovered Class	Conf.	Declared RoPA Purpose	Retention	Status
customer_signups.csv **customer_name**	Personal identity	**100%**	Customer Order Fulfillment & Shipping	36m (Account Deletion)	**ACTIVE**
customer_signups.csv **email_address**	Contact information	**100%**	Customer Order Fulfillment & Shipping	36m (Account Deletion)	**ACTIVE**
customer_signups.csv **phone_number**	Contact information	**100%**	Customer Order Fulfillment & Shipping	24m (Order Completion)	**ACTIVE**
customer_signups.csv **date_of_birth**	Personal identity	**100%**	*Missing Purpose*	*Missing*	**UNREGISTERED**
customer_signups.csv **gender**	Personal identity	**100%**	*Missing Purpose*	12m (Account Review)	**UNLINKED_PURPOSE**
customer_signups.csv **residential_address**	Location / address	**100%**	Customer Order Fulfillment & Shipping	*Missing*	**INCOMPLETE**
customer_signups.csv **pin_code**	Location / address	**100%**	Customer Order Fulfillment & Shipping	36m (Account Deletion)	**ACTIVE**
digital_payments_log.csv **bank_account_number**	Financial information	**100%**	Payment Processing & Financial Ledger	84m (Financial Year End)	**ACTIVE**
digital_payments_log.csv **ifsc_code**	Financial information	**100%**	Payment Processing & Financial Ledger	84m (Financial Year End)	**ACTIVE**
digital_payments_log.csv **card_number**	Financial information	**100%**	*Missing Purpose*	24m (Card Expiration)	**UNLINKED_PURPOSE**
digital_payments_log.csv **upi_id**	Financial information	**100%**	*Missing Purpose*	*Missing*	**UNREGISTERED**
kyc_verification_records.csv **aadhaar_number**	Government identifiers	**100%**	Statutory KYC & Customer Verification	60m (Statutory Audit Window)	**ACTIVE**
kyc_verification_records.csv **pan_number**	Government identifiers	**100%**	Statutory KYC & Customer Verification	60m (Statutory Audit Window)	**ACTIVE**
kyc_verification_records.csv **passport_number**	Government identifiers	**100%**	Statutory KYC & Customer Verification	*Missing*	**INCOMPLETE**
kyc_verification_records.csv **voter_id**	Government identifiers	**100%**	*Missing Purpose*	*Missing*	**UNREGISTERED**
kyc_verification_records.csv **driving_licence**	Government identifiers	**100%**	*Missing Purpose*	*Missing*	**UNREGISTERED**
enterprise_data.db (employees) **employee_name**	Personal identity	**100%**	Employee Payroll & Statutory Benefits	84m (Termination of Employment)	**ACTIVE**
enterprise_data.db (employees) **age**	Personal identity	**100%**	Employee Payroll & Statutory Benefits	84m (Termination of Employment)	**ACTIVE**
enterprise_data.db (employees) **pan_card**	Government identifiers	**100%**	Employee Payroll & Statutory Benefits	84m (Statutory Tax Filing)	**ACTIVE**
enterprise_data.db (employees) **salary_account_no**	Financial information	**100%**	Employee Payroll & Statutory Benefits	84m (Termination of Employment)	**ACTIVE**
enterprise_data.db (employees) **bank_ifsc**	Financial information	**100%**	Employee Payroll & Statutory Benefits	84m (Termination of Employment)	**ACTIVE**
enterprise_data.db (delivery_dispatch) **delivery_address**	Location / address	**100%**	Customer Order Fulfillment & Shipping	12m (Delivery Confirmation)	**ACTIVE**
enterprise_data.db (delivery_dispatch) **postal_pin**	Location / address	**100%**	Customer Order Fulfillment & Shipping	12m (Delivery Confirmation)	**ACTIVE**

Target Asset / Column	Discovered Class	Conf.	Declared RoPA Purpose	Retention	Status
enterprise_data.db (delivery_dispatch) **geo_coords**	Location / address	**100%**	Customer Order Fulfillment & Shipping	*Missing*	**INCOMPLETE**
enterprise_data.db (delivery_dispatch) **order_upi**	Financial information	**100%**	Payment Processing & Financial Ledger	24m (Dispute Window)	**ACTIVE**
enterprise_data.db (legacy_unindexed_vault) **aadhaar_card**	Government identifiers	**100%**	*Missing Purpose*	*Missing*	**UNREGISTERED**
enterprise_data.db (legacy_unindexed_vault) **caller_phone**	Contact information	**100%**	*Missing Purpose*	*Missing*	**UNREGISTERED**
enterprise_data.db (legacy_unindexed_vault) **contact_email**	Contact information	**100%**	*Missing Purpose*	*Missing*	**UNREGISTERED**
enterprise_data.db (legacy_unindexed_vault) **card_number**	Financial information	**100%**	*Missing Purpose*	*Missing*	**UNREGISTERED**
enterprise_data.db (legacy_unindexed_vault) **birth_date**	Personal identity	**100%**	*Missing Purpose*	*Missing*	**UNREGISTERED**
enterprise_data.db (legacy_unindexed_vault) **geo_coordinates**	Location / address	**100%**	*Missing Purpose*	*Missing*	**UNREGISTERED**

3. Evidence Gaps & Missing Metadata Action Checklist

Sev.	Target Field Identifier	Gap Type	Evidence Findings & Recommended Human Action
CRITICAL	`customer_signups.csv::date_of_birth`	UNREGISTERED_SHADOW_DATA	**Unregistered Candidate PII in customer_signups.csv**: Field 'date_of_birth' was identified as candidate personal data (Personal identity, confidence: 100%) across 150 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning.
CRITICAL	`digital_payments_log.csv::upi_id`	UNREGISTERED_SHADOW_DATA	**Unregistered Candidate PII in digital_payments_log.csv**: Field 'upi_id' was identified as candidate personal data (Financial information, confidence: 100%) across 100 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning.
CRITICAL	`kyc_verification_records.csv::voter_id`	UNREGISTERED_SHADOW_DATA	**Unregistered Candidate PII in kyc_verification_records.csv**: Field 'voter_id' was identified as candidate personal data (Government identifiers, confidence: 100%) across 120 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning.
CRITICAL	`kyc_verification_records.csv::driving_licence`	UNREGISTERED_SHADOW_DATA	**Unregistered Candidate PII in kyc_verification_records.csv**: Field 'driving_licence' was identified as candidate personal data (Government identifiers, confidence: 100%) across 120 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning.
CRITICAL	`enterprise_data.db:legacy_unindexed_vault::aadhaar_card`	UNREGISTERED_SHADOW_DATA	**Unregistered Candidate PII in enterprise_data.db:legacy_unindexed_vault**: Field 'aadhaar_card' was identified as candidate personal data (Government identifiers, confidence: 100%) across 70 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning.
CRITICAL	`enterprise_data.db:legacy_unindexed_vault::caller_phone`	UNREGISTERED_SHADOW_DATA	**Unregistered Candidate PII in enterprise_data.db:legacy_unindexed_vault**: Field 'caller_phone' was identified as candidate personal data (Contact information, confidence: 100%) across 70 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning.
CRITICAL	`enterprise_data.db:legacy_unindexed_vault::contact_email`	UNREGISTERED_SHADOW_DATA	**Unregistered Candidate PII in enterprise_data.db:legacy_unindexed_vault**: Field 'contact_email' was identified as candidate personal data (Contact information, confidence: 100%) across 70 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning.

Sev.	Target Field Identifier	Gap Type	Evidence Findings & Recommended Human Action
CRITICAL	`enterprise_data.db:legacy_unindexed_vault::card_number`	UNREGISTERED_SHADOW_DATA	**Unregistered Candidate PII in enterprise_data.db:legacy_unindexed_vault**: Field 'card_number' was identified as candidate personal data (Financial information, confidence: 100%) across 70 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning.
CRITICAL	`enterprise_data.db:legacy_unindexed_vault::birth_date`	UNREGISTERED_SHADOW_DATA	**Unregistered Candidate PII in enterprise_data.db:legacy_unindexed_vault**: Field 'birth_date' was identified as candidate personal data (Personal identity, confidence: 100%) across 70 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning.
CRITICAL	`enterprise_data.db:legacy_unindexed_vault::geo_coordinates`	UNREGISTERED_SHADOW_DATA	**Unregistered Candidate PII in enterprise_data.db:legacy_unindexed_vault**: Field 'geo_coordinates' was identified as candidate personal data (Location / address, confidence: 100%) across 70 records, but has NO entry in the governance registry. ■ **Action:** Human Review Required: Consult data owner to verify business necessity. If legitimate, register processing purpose and retention policy in RoPA. If obsolete, initiate data minimization or decommissioning.
HIGH	`customer_signups.csv::gender`	MISSING_PURPOSE	**Missing Declared Purpose for gender**: Field 'gender' is registered under customer_signups.csv but has no declared business processing purpose. ■ **Action:** Privacy Steward Action: Assign and validate a declared business purpose prior to lawful processing verification.
HIGH	`customer_signups.csv::residential_address`	MISSING_RETENTION	**Missing Retention Schedule for residential_address**: Field 'residential_address' contains candidate personal data without a defined retention limit or deletion trigger. ■ **Action:** Data Governance Action: Formulate data retention schedule and disposal trigger in accordance with corporate data retention policy.
HIGH	`digital_payments_log.csv::card_number`	MISSING_PURPOSE	**Missing Declared Purpose for card_number**: Field 'card_number' is registered under digital_payments_log.csv but has no declared business processing purpose. ■ **Action:** Privacy Steward Action: Assign and validate a declared business purpose prior to lawful processing verification.
HIGH	`kyc_verification_records.csv::passport_number`	MISSING_RETENTION	**Missing Retention Schedule for passport_number**: Field 'passport_number' contains candidate personal data without a defined retention limit or deletion trigger. ■ **Action:** Data Governance Action: Formulate data retention schedule and disposal trigger in accordance with corporate data retention policy.
HIGH	`enterprise_data.db:delivery_dispatch::geo_coords`	MISSING_RETENTION	**Missing Retention Schedule for geo_coords**: Field 'geo_coords' contains candidate personal data without a defined retention limit or deletion trigger. ■ **Action:** Data Governance Action: Formulate data retention schedule and disposal trigger in accordance with corporate data retention policy.

4. Benchmark Precision & Recall Performance

Evaluated across **44 fields** against ground truth. Macro Precision: **100.0%** | Macro Recall: **100.0%** | Macro F1-Score: **100.0%**

Code	Predefined Data Class	Support	TP	FP	FN	Precision	Recall	F1-Score
PID	Personal identity	6	6	0	0	100.0%	100.0%	**100.0%**
CNT	Contact information	4	4	0	0	100.0%	100.0%	**100.0%**
GID	Government identifiers	7	7	0	0	100.0%	100.0%	**100.0%**
FIN	Financial information	8	8	0	0	100.0%	100.0%	**100.0%**
LOC	Location / address	6	6	0	0	100.0%	100.0%	**100.0%**